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Submission Topic Assignment 2

Due date of submission 10/09/2020

Due time of submission 1:00 PM

Group 05

Actual submission date _____

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Assignment 2

Q1. (a) $7260 = 2^2 \times 3 \times 5 \times 11^2$

$5720 = 2^3 \times 5 \times 11 \times 13$

$\gcd(7260, 5720) = 2^2 \times 5 \times 11$

$\text{lcm}(7260, 5720) = 2^3 \times 3 \times 5 \times 11^2 \times 13 = 188760$

(b) $434346 = 43095 \times 2 + 18156$ $\gcd(434346, 43095) = \gcd(43095, 18156)$

$43095 = 18156 \times 2 + 6783$ $\gcd(43095, 18156) = \gcd(18156, 6783)$

$18156 = 6783 \times 2 + 4590$ $\gcd(18156, 6783) = \gcd(6783, 4590)$

$6783 = 4590 \times 1 + 2193$ $\gcd(6783, 4590) = \gcd(4590, 2193)$

$4590 = 2193 \times 2 + 204$ $\gcd(4590, 2193) = \gcd(2193, 204)$

$2193 = 204 \times 10 + 153$ $\gcd(2193, 204) = \gcd(204, 153)$

$204 = 153 \times 1 + 51$ $\gcd(204, 153) = \gcd(153, 51)$

$153 = 51 \times 3 + 0$ $\gcd(153, 51) = \gcd(51, 0) = 51$.

True.

Q2. (a) Proof: $\because d \mid a$

$\therefore a = dk$ for $k \in \mathbb{Z}$.

$\because d \mid b$

$\therefore b = dx$ for $x \in \mathbb{Z}$.

$$\frac{ab}{d} = \frac{dk \cdot dx}{d}$$

$$= \frac{d^2 kx}{d}$$

$= dkx$ since $d \in \mathbb{N}$

$= ax$ or bk

$\therefore \frac{ab}{d}$ is an integer that is a common multiple of both a and b. □

(b) False.

Take $a = 4, b = 2, d = 6 \in \mathbb{N}$.

(a) Then $a^2 = 4^2 = 16$, $b^2 = 2^2 = 4$

$\because$ $16 \bmod 6 = 4$, $4 \bmod 6 = 4 \in \mathbb{N}$

$\therefore$ $16 \equiv 4 \pmod 6$

$\therefore$ $a^2 \equiv b^2 \pmod 6$

Also $\because$ $4 \bmod 6 = 4$, $2 \bmod 6 = 2$

$\therefore$ $4 \not\equiv 2 \pmod 6$.

$\therefore$ $a \not\equiv b \pmod 6$. $\square$

(c) True.

Proof (by contradiction):

Suppose this statement is false, there is, $\exists a, b \in \mathbb{N}$, $a^3 \equiv b^3 \pmod 3 \rightarrow a \not\equiv b \pmod 3$.

Then, $a^3 - b^3 = 3k$ for some $k \in \mathbb{Z}$

So $(a-b)(a^2+b^2+ab) = 3k$

Also $\because$ $a \not\equiv b \pmod 3$

$\therefore$ $a - b \neq 3x$ for some $x \in \mathbb{Z}$

$\therefore$ $a^2 + b^2 + ab = 3y$ for some $y \in \mathbb{Z}$.

There are 2 possibilities for a, b such that $a - b \neq 3x$.

① $(a-b) \equiv 1 \pmod 3$

$a \equiv 1 \pmod 3$, $b \equiv 0 \pmod 3$ or vice versa

$\therefore$ $a^2 + b^2 + ab \equiv 1 \pmod 3$

$\therefore$ $a^2 + b^2 + ab \neq 3y$

So we have a contradiction.

② $(a-b) \equiv 2 \pmod 3$

$\therefore$ $a \equiv 0 \pmod 3$, $b \equiv 2 \pmod 3$ or vice versa

$\therefore$ $a^2 + b^2 + ab \equiv 4 \equiv 1 \pmod 3$

$\therefore$ $a^2 + b^2 + ab \neq 3y$

So we have a contradiction.

Therefore, for all $a, b \in \mathbb{N}$, if $a^3 \equiv b^3 \pmod 3$, then $a \equiv b \pmod 3$ $\square$

(d) True.

Proof (by contraposition).

Suppose that $\sqrt{x}$ is not irrational, namely rational.

$\therefore$ $\sqrt{x} = \frac{a}{b}$ where $a, b \in \mathbb{N}$, $\gcd(a, b) = 1$

$\therefore (\sqrt{x})^2 = \frac{a^2}{b^2}$

$x = \frac{a^2}{b^2}$

$\therefore x$ is rational number

Therefore, by contraposition, we conclude that For all $x \in \mathbb{R}$, if x is irrational and $x \geq 0$, then $\sqrt{x}$ is also irrational. $\square$

(e). False.

Take $x = \frac{16}{25}$

Then $x \geq 0$ and $x \notin \mathbb{Z}$

But $\sqrt{x} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ is rational. $\square$

(f) True.

Proof: $\because x \leq y$, $\lfloor x \rfloor = \lfloor y \rfloor$

$\therefore x = a + b$, $y = a + c$ where $a \in \mathbb{N}$, $b, c \in [0, 1)$, $b \leq c$

$\therefore 2x = 2(a+b) = 2a + 2b$

$2y = 2(a+c) = 2a + 2c$

There are 3 possibilities for $\lfloor x \rfloor = \lfloor y \rfloor$.

① $b \in [0, \frac{1}{2})$, $c \in [\frac{1}{2}, 1)$

Then $\lfloor 2x \rfloor = 2a$, $\lfloor 2y \rfloor = 2a$

$\therefore \lfloor 2x \rfloor = \lfloor 2y \rfloor$

② $b \in [0, \frac{1}{2})$, $c \in [\frac{1}{2}, 1)$

Then $\lfloor 2x \rfloor = 2a$, $\lfloor 2y \rfloor = 2a + 1$

$\therefore \lfloor 2x \rfloor = \lfloor 2y \rfloor - 1$

③ $b \in [\frac{1}{2}, 1)$, $c \in [\frac{1}{2}, 1)$

Then $\lfloor 2x \rfloor = 2a + 1$, $\lfloor 2y \rfloor = 2a + 1$

$\therefore \lfloor 2x \rfloor = \lfloor 2y \rfloor$

Therefore, for all $x, y \in \mathbb{R}$, if $x \leq y$ and $\lfloor x \rfloor = \lfloor y \rfloor$, the either $\lfloor 2x \rfloor = \lfloor 2y \rfloor$ or $\lfloor 2x \rfloor = \lfloor 2y \rfloor - 1$. $\square$

Q3. Proof (by contradiction): Suppose $\sqrt{10}$ is rational.

Then $\sqrt{10} = \frac{a}{b}$ where $a \in \mathbb{W}$, $b \in \mathbb{N}$, $\gcd(a, b) = 1$

$\therefore (\sqrt{10})^2 = \frac{a^2}{b^2}$

$10 = \frac{a^2}{b^2}$

$a^2 = 10b^2$

$a^2 = 2 \times 5b^2$

$\therefore a^2$ is an even number

Also $\because$ odd number times odd number is always odd.

$\therefore a$ is also an even number.

$\therefore a = 2x$ for some $x \in \mathbb{W}$

$\therefore (2x)^2 = 2 \cdot 5b^2$

$4x^2 = 10b^2$

$2x^2 = 5b^2$

$\because 2x^2$ is even, 5 is odd

$\therefore b^2$ must be even.

$\therefore b$ must be even.

Hence, both a, b are even

$\therefore \gcd(a, b) = 2 \neq 1$

So we have a contradiction

Therefore $\sqrt{10}$ cannot be rational, that is $\sqrt{10}$ is irrational. $\square$